

W11D1 - Raccolta Informazioni

metasploitable - raccolta informazioni

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Sommario

Con l'esercizio di oggi andremo a testare la macchina Metasploitable (192.168.50.101) alla ricerca di informazioni utili per un successivo Penetration Testing.

I test effettuati comprenderanno la rilevazione del sistema operativo in uso sulla macchina target, una scansione SYN e una TCP oltre che una scansione delle versioni dei servizi eventualmente in ascolto sulle porte rilevate.

Le Due macchine si trovano su sottoreti diverse ma sono in grado di dialogare tra loro come si evince dagli screenshot seguenti:

- **Kali Linux (attaccante)**

```
Session Actions Edit View Help
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
   inet 192.168.50.100/24 brd 192.168.50.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
   inet6 fe80::89c3:47eb:da38:43e0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:98:9a:fd brd ff:ff:ff:ff:ff:ff
(kali@kali)-[~]
$
```

- Metasploitable (target)

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:b5:7a:73 brd ff:ff:ff:ff:ff:ff
    inet 192.168.51.101/24 brd 192.168.51.255 scope global eth0
    inet6 fe80::a00:27ff:feb5:7a73/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ _
```

- Ping test

```
(kali㉿kali)-[~]
$ ping -c 4 metasploit-nokali
PING metasploit-nokali (192.168.51.101) 56(84) bytes of data.
64 bytes from metasploit-nokali (192.168.51.101): icmp_seq=1 ttl=63 time=0.274 ms
64 bytes from metasploit-nokali (192.168.51.101): icmp_seq=2 ttl=63 time=0.270 ms
64 bytes from metasploit-nokali (192.168.51.101): icmp_seq=3 ttl=63 time=0.278 ms
64 bytes from metasploit-nokali (192.168.51.101): icmp_seq=4 ttl=63 time=0.238 ms

— metasploit-nokali ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.238/0.265/0.278/0.015 ms
```

Oggetto del test (scope)

La macchina oggetto dei nostri test sarà la metasploitable all'indirizzo:

- 192.168.50.101

Risultati in breve

#	Porte Rilevate	Stato	Servizio	Descrizione
1	21/tcp	open	ftp	file transfer protocol
2	22/tcp	open	ssh	secure shell
3	23/tcp	open	telnet	client-server network protocol
4	25/tcp	open	smtp	simple mail transfer protocol
5	53/tcp	open	domain	DNS (domain name system) protocol
6	80/tcp	open	http	web server
7	111/tcp	open	rpcbind	RPC (remote procedure call) protocol
8	139/tcp	open	netbios-ssn	Netbios session protocol
9	445/tcp	open	microsoft-ds	SMB (service message block) protocol

#	Porte Rilevate	Stato	Servizio	Descrizione
10	512/tcp	open	exec	Rexec (remote exec)
11	513/tcp	open	login	Rlogin (remote login)
12	514/tcp	open	shell	Rsh (remote shell)
13	1099/tcp	open	rmiregistry	Java RMI registry (part of Java Remote Method Invocation)
14	1524/tcp	open	ingreslock	Ingres SQL database
15	2049/tcp	open	nfs	Network File System protocol
16	2121/tcp	open	ccproxy-ftp	part of FTP
17	3306/tcp	open	mysql	mysql database
18	5432/tcp	open	postgresql	postgresql database
19	5900/tcp	open	vnc	Virtual Network Computing (a remote desktop protocol)
20	6000/tcp	open	X11	X window system
21	6667/tcp	open	irc	Internet Relay Chat protocol
22	8009/tcp	open	ajp13	Apache JServ protocol (an optimized version of the http protocol)
23	8180/tcp	open	unknown	Apache Tomcat (a Java http web server)

Queste sono le porte aperte e i relativi servizi in ascolto, trovati con le nostre scansioni sul target.

Tecnologie utilizzate

Per questi test abbiamo utilizzato il seguente software:

- nmap 7.95

Risultati

⇒ OS Rilevato

Descrizione

L'OS rilevato durante le nostre scansioni risulta essere **Linux**, con un kernel tra i **2.6.15** e **2.6.26**.

```
(kali㉿kali)-[~]
$ nmap -O metasploit-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 10:47 EDT
Nmap scan report for metasploit-nokali (192.168.51.101)
Host is up (0.00048s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.15 - 2.6.26 (likely embedded)
Network Distance: 2 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.59 seconds

(kali㉿kali)-[~]
```

⇒ SYN Scan

Descrizione

```
(kali㉿kali)-[~]  
$ nmap -sS metasploit-nokali  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 11:07 EDT  
Nmap scan report for metasploit-nokali (192.168.51.101)  
Host is up (0.0013s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds
```

⇒ TCP Connect Scan

Descrizione

```
(kali㉿kali)-[~]  
$ nmap -sT metasploit-nokali  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 11:09 EDT  
Nmap scan report for metasploit-nokali (192.168.51.101)  
Host is up (0.00052s latency).  
Not shown: 977 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

⇒ Services Version Detection

Descrizione

```
(kali@kali)~[~]
$ nmap -sV metasploit-nokali
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-22 11:11 EDT
Nmap scan report for metasploit-nokali (192.168.51.101)
Host is up (0.00060s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 52.30 seconds
```

Tutte le scansioni hanno riportato gli stessi servizi in ascolto e le medesime porte. Chiaramente la scansione più lunga è stata l'ultima in quanto ha eseguito diversi script per estrarre le versioni dei servizi in ascolto.

Con questo si conclude il nostro esame sulla macchina Metasploitable2.

Alla prossima.

